

Question #1 of 23

Question ID: 1472207

The technique in which a machine learns to model a set of output data from a given set of inputs is *best* described as:

- A) unsupervised learning.
- B) deep learning.
- C) supervised learning.



Explanation

Supervised learning is a machine learning technique in which a machine is given labelled input and output data and models the output data based on the input data. In unsupervised learning, a machine is given input data in which to identify patterns and relationships, but no output data to model. Deep learning is a technique to identify patterns of increasing complexity and may use supervised or unsupervised learning.

(Module 3.1, LOS 3.a)

Hanna Kowalski is a senior fixed-income portfolio analyst at Czarnaskala BP. Kowalski supervises Lena Nowak, who is a junior analyst.

Over the past several years, Kowalski has become aware that investment firms are increasingly using technology to improve their investment decision making. Kowalski has become particularly interested in machine learning techniques and how they might be applied to investment management applications.

Kowalski has read a number of articles about machine learning in various journals for financial analysts. However, she has only a minimal knowledge of how she might source appropriate model inputs, interpret model outputs, and translate those outputs into investment actions.

Kowalski and Nowak meet to discuss plans for incorporating machine learning into their investment model. Kowalski asks Nowak to research machine learning and report back on the types of investment problems that machine learning can address, how the algorithms work, and what the various terminology means.

After spending a few hours researching the topic, Nowak makes a number of statements to Kowalski on the topics of:

- Classification and regression trees (CART)
- Hierarchical clustering

- Neural networks
- Reinforcement learning (RL) algorithms

Kowalski is left to work out which of Nowak's statements are fully accurate and which are not.

Question #2 - 5 of 23

Question ID: 1472228

Nowak first tries to explain classification and regression tree (CART) to Kowalski. CART is *least* likely to be applied to predict a:

- A) continuous target variable, producing a regression tree.
- B) categorical target variable, producing a classification tree.
- C) discrete target variable, producing a cardinal tree.



Explanation

Classification and regression trees (CART) are generally applied to predict either a continuous target variable, producing a regression tree, or a categorical target variable, producing a classification tree.

(Module 3.2, LOS 3.c)

Question #3 - 5 of 23

Question ID: 1472229

Which of the following statements Nowak makes about hierarchical clustering is *most* accurate?

- A) Bottom-up hierarchical clustering begins with each observation being its own cluster.
- B) In divisive hierarchical clustering, the algorithm seeks out the two closest clusters.
- C) Hierarchical clustering is a supervised iterative algorithm that is used to build a hierarchy of clusters.



Explanation

Agglomerative (bottom-up) hierarchical clustering begins with each observation being its own cluster. Then, the algorithm finds the two closest clusters, and combines them into a new, larger cluster. Hierarchical clustering is an unsupervised iterative algorithm. Divisive (top-down) hierarchical clustering progressively partitions clusters into smaller clusters until each cluster contains only one observation.

(Module 3.3, LOS 3.d)

Question #4 - 5 of 23

Question ID: 1472230

Which of the following statements Nowak makes about neural networks is *most* accurate?

Neural networks:

- A) have four types of layers: an input layer, agglomerative layers, regularization layers, and an output layer. 
- B) have an input layer node that consists of a summation operator and an activation function. 
- C) are effective in tasks with non-linearities and complex interactions among variables. 

Explanation

Neural networks have been successfully applied to solve a variety of problems characterized by non-linearities and complex interactions among variables. Neural networks have three types of layers: an input layer, hidden layers, and an output layer. The hidden layer nodes (not the input layer nodes) each consist of a summation operator and an activation function; these nodes are where learning takes place.

(Module 3.3, LOS 3.e)

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Question ID: 1472231

Nowak tries to explain the reinforcement learning (RL) algorithm to Kowalski and makes a number of statements about it. The reinforcement learning (RL) algorithm involves an agent that is *most* likely to:

- A) take into consideration the constraints of its environment. 
- B) perform actions that will minimize costs over time. 
- C) make use of direct labeled data and instantaneous feedback. 

Explanation

The reinforcement learning (RL) algorithm involves an agent that will perform actions that will maximize its rewards over time, taking into consideration the constraints of the environment. Unlike supervised learning, reinforcement learning has neither instantaneous feedback nor direct labeled data for each observation.

(Module 3.3, LOS 3.e)

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Question ID: 1472220

The unsupervised machine learning algorithm that reduces highly correlated features into fewer uncorrelated composite variables by transforming the feature covariance matrix *best* describes:

A) principal components analysis



B) k-means clustering



C) hierarchical clustering



Explanation

Principal components analysis (PCA) is an unsupervised machine learning algorithm that reduces highly correlated features into fewer uncorrelated composite variables by transforming the feature covariance matrix. K-means partitions observations into a fixed number (k) of non-overlapping clusters. Hierarchical clustering is an unsupervised iterative algorithm used to build a hierarchy of clusters.

(Module 3.3, LOS 3.d)

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Question ID: 1472216

Considering the various supervised machine learning algorithms, a penalized regression where the penalty term is the sum of the absolute values of the regression coefficients best describes:

A) support vector machine (SVM).



B) k-nearest neighbor (KNN).



C) least absolute shrinkage and selection operator (LASSO)



Explanation

LASSO (least absolute shrinkage and selection operator) is a popular type of penalized regression in which the penalty term comprises summing the absolute values of the regression coefficients. The more included features, the larger the penalty will be. The result is that a feature needs to make a sufficient contribution to model fit to offset the penalty from including it.

(Module 3.2, LOS 3.c)

Joyce Tan manages a medium-sized investment fund at Marina Bay Advisors that specializes in international large cap equities. Over the four years that she has been portfolio manager, Tan has been invested in approximately 40 stocks at a time.

Tan has used a number of methodologies to select investment opportunities from the universe of investable stocks. In some cases, Tan uses quantitative measures such as accounting ratios to find the most promising investment candidates. In other cases, her team of analysts suggest investments based on qualitative factors and various investment hypotheses.

Tan begins to wonder if her team could leverage financial technology to make better decisions. Specifically, she has read about various machine learning techniques to extract useful information from large financial datasets, in order to uncover new sources of alpha.

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Question ID: 1472223

Tan is interested in using a supervised learning algorithm to analyze stocks. This task is *least* likely to be a classification problem if the target variable is:

A) ordinal.



B) continuous.



C) categorical.



Explanation

Supervised learning can be divided into two categories: regression and classification. If the target variable is categorical or ordinal (e.g., determining a firm's rating), then it is a classification problem. If the target variable to be predicted is continuous, then the task is one of regression.

(Module 3.1, LOS 3.a)

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Question ID: 1472224

After Tan implements a particular new supervised machine learning algorithm, she begins to suspect that the holdout samples she is using are reducing the training set size too much. As a result, she begins to make use of K-fold cross-validation. In the K-fold cross-validation technique, after Tan shuffles the data randomly it is most likely that:

A) the data will be divided into $k - 1$ equal sub-samples.



B) $k - 1$ samples will be used as validation samples.



C) $k - 1$ samples will be used as training samples.



Explanation

In the K-fold cross-validation technique, the data is shuffled randomly and then divided into k equal sub-samples. One sample is saved to be used as a validation sample, and the other $k - 1$ samples are used as training samples.

(Module 3.1, LOS 3.b)

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Question ID: 1472225

At first Tan bases her stock picks on the results of a single machine-learning model, but then begins to wonder if she should instead be using the predictions of a group of models. Compared to a single machine-learning model, an ensemble machine learning algorithm is *most* likely to produce predictions that are:

A) more precise but less dependable.



B) more accurate and more stable.



C) less reliable but more steady.



Explanation

Ensemble learning, which is a technique of combining the predictions from a number of models, generally results in more accurate and more stable predictions than a single model.

(Module 3.2, LOS 3.c)

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Question ID: 1472226

Tan is interested in applying neural networks, deep learning nets, and reinforcement learning to her investment process. Regarding these techniques, which of the following statements is *most* accurate?

- A) Neural networks with one or more hidden layers would be considered deep learning nets (DLNs). ✗
- B) Reinforcement learning algorithms achieve maximum performance when they stay as far away from their constraints as possible. ✗
- C) Neural networks work well in the presence of non-linearities and complex interactions among variables. ✓

Explanation

Neural networks have been successfully applied to a variety of investment tasks characterized by non-linearities and complex interactions among variables.

Neural networks with at least three hidden layers are known as deep learning nets (DLNs). Reinforcement learning algorithms use an agent that will maximize its rewards over time, within the constraints of its environment.

(Module 3.3, LOS 3.e)

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Question ID: 1472210

Which supervised learning model is *most appropriate* (1) when the Y-variable is continuous and (2) when the Y-variable is categorical

Continuous Y-variable Categorical Y-variable

- | | | |
|-------------------|-----------------|---|
| A) Regression | Classification | ✓ |
| B) Decision trees | Regression | ✗ |
| C) Classification | Neural Networks | ✗ |

Explanation

When the Y-variable is continuous, the appropriate approach is that of regression (used in a broad, ML context). When the Y-variable is categorical (i.e., belonging to a category or classification) or ordinal (i.e., ordered or ranked), a classification model is used.

(Module 3.1, LOS 3.a)

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Question ID: 1472209

Which of the following about unsupervised learning is *most accurate*?

- A) Unsupervised learning has lower forecasting accuracy as compared to supervised learning. 
- B) Classification is an example of unsupervised learning algorithm. 
- C) There is no labeled data. 

Explanation

In unsupervised learning, the ML program is not given labeled training data. Instead, inputs are provided without any conclusions about those inputs. In the absence of any tagged data, the program seeks out structure or inter-relationships in the data. Clustering is one example of the output of unsupervised ML program while classification is suited for supervised learning.

(Module 3.1, LOS 3.a)

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Question ID: 1472218

What is the appropriate remedy in the presence of excessive number of features in a data set?

- A) Dimension reduction. 
- B) Unsupervised learning. 
- C) Big data analysis. 

Explanation

Big Data refers to very large data sets which may include both structured (e.g. spreadsheet) data and unstructured (e.g. emails, text, or pictures) data and includes a large number of features as well as number of observations. Dimension reduction seeks to remove the noise (i.e., those attributes that do not contain much information) when the number of features in a data set (its dimension) is excessive.

(Module 3.3, LOS 3.d)

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Question ID: 1472213

The degree to which a machine learning model retains its explanatory power when predicting out-of-sample is *most* commonly described as:

- A) hegemony. 
- B) generalization. 

C) predominance.



Explanation

Generalization describes the degree to which, when predicting out-of-sample, a machine learning model retains its explanatory power.

(Module 3.1, LOS 3.b)

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Question ID: 1472217

Considering the various supervised machine learning algorithms, a linear classifier that seeks the optimal hyperplane and is typically used for classification, best describes:

A) support vector machine (SVM).



B) k-nearest neighbor (KNN).



C) classification and regression tree (CART)



Explanation

Support vector machine (SVM) is a linear classifier that aims to seek the optimal hyperplane, i.e. the one that separates the two sets of data points by the maximum margin. SVM is typically used for classification.

(Module 3.2, LOS 3.c)

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Question ID: 1472212

Overfitting is *least likely* to result in:

A) inclusion of noise in the model.



B) higher number of features included in the data set.



C) higher forecasting accuracy in out-of-sample data.



Explanation

Overfitting results when a large number of features (i.e., independent variables) are included in the data sample. The resulting model can use the "noise" in the dependent variables to improve the model fit. Overfitting the model in this way will actually decrease the accuracy of model forecasts on other (out-of-sample) data.

(Module 3.1, LOS 3.b)

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Question ID: 1472208

Which of the following statements about supervised learning is *most accurate*?

- A) Supervised learning requires human intervention in machine learning process. 
- B) Typical data analytics tasks for supervised learning include classification and prediction. 
- C) Supervised learning does not differentiate between tag and features. 

Explanation

Supervised learning utilizes labeled training data to guide the ML program but does not need "human intervention." Typical data analytics tasks for supervised learning include classification and prediction.

(Module 3.1, LOS 3.a)

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Question ID: 1472219

Dimension reduction is *most likely* to be an example of:

- A) unsupervised learning. 
- B) clustering. 
- C) supervised learning. 

Explanation

Dimension reduction and clustering are examples of unsupervised learning algorithms.

(Module 3.3, LOS 3.d)

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Question ID: 1472211

A rudimentary way to think of machine learning algorithms is that they:

- A) "synthesize the pattern, review the pattern." 
- B) "develop the pattern, interpret the pattern." 
- C) "find the pattern, apply the pattern." 

Explanation

One elementary way to think of ML algorithms is to "find the pattern, apply the pattern." Machine learning attempts to extract knowledge from large amounts of data by learning from known examples in order to determine an underlying structure in the data. The focus is on generating structure or predictions without human intervention.

(Module 3.1, LOS 3.a)

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Question ID: 1472221

An algorithm that involves an agent that performs actions that will maximize its rewards over time, taking into consideration the constraints of its environment, *best* describes:

- A) deep learning nets.
- B) reinforcement learning.
- C) neural networks.



Explanation

Reinforcement learning algorithms involve an agent that will perform actions that will maximize its rewards over time, taking into consideration the constraints of its environment. Neural networks consist of nodes connected by links; learning takes place in the hidden layer nodes, each of which consists of a summation operator and an activation function. Neural networks with many hidden layers (often more than 20) are known as deep learning nets (DLNs) and used in artificial intelligence.

(Module 3.3, LOS 3.e)

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Question ID: 1472215

A random forest is *least likely* to:

- A) provide a solution to overfitting problem.
- B) be a classification tree.
- C) reduce signal-to-noise ratio.



Explanation

Random forest is a collection of randomly generated classification trees from the same data set. A randomly selected subset of features is used in creating each tree and hence each tree is slightly different from the others. Since each tree only uses a subset of features, random forests can mitigate the problem of overfitting. Because errors across different trees tend to cancel each other out, using random forests can increase the signal-to-noise ratio.

(Module 3.2, LOS 3.c)

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Question ID: 1472214

In machine learning, out-of-sample error equals:

- A) bias error plus variance error plus base error. 
- B) Standard error plus data error plus prediction error. 
- C) forecast error plus expected error plus regression error. 

Explanation

Out-of-sample error equals bias error plus variance error plus base error. Bias error is the extent to which a model fits the training data. Variance error describes the degree to which a model's results change in response to new data from validation and test samples. Base error comes from randomness in the data.

(Module 3.1, LOS 3.b)